

PRAYAG RANJAN SAHU

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CURRENT AFFILIATION

Research Assistant

October 2025 – present

Advisor: Prof. Jer-Lai Kuo

Institute of Atomic and Molecular Sciences (IAMS), Academia Sinica, Taiwan

EDUCATION

Integrated Master's in Chemistry | CGPA: 9.45/10

Dec. 2020 – May 2025

Advisor: Prof. Upakarasamy Lourderaj

National Institute of Science Education and Research (NISER), Bhubaneswar

AWARDS & SCHOLARSHIPS

Institute Gold Medal

July 2025

Best Overall Outstanding Academic Performance in the Class of 2020 across all Schools

NISER, Bhubaneswar

School Silver Medal

July 2025

Outstanding Academic Performance in the School of Chemical Sciences in the Class of 2020

NISER, Bhubaneswar

Smt. Jayalaxamma Memorial Award in Chemistry

July 2025

Highest CGPA in the School of Chemical Sciences in the Class of 2020

NISER, Bhubaneswar

Academic Excellence Awards for Semesters IV, V, VI, VII

2022 - 2024

Highest SGPA in the Class of 2020 and/or the School of Chemical Sciences

NISER, Bhubaneswar

International Internship Program (IIP) Scholarship

May-July 2024

Summer Internship Scholarship for International Students

IAMS, Taiwan

DISHA Scholarship

2020 - 2025

DAE-Integrated Scholarship for Higher Education and Advancement

Dept. of Atomic Energy (DAE), Gov. of India

RESEARCH EXPERIENCE

Machine Learning Studies on the Thermal Denitrogenation of 1-Pyrazoline

Aug. 2023 – July 2025

Master's Thesis - Supervised by Prof. Upakarasamy Lourderaj

NISER, Bhubaneswar

- Developed a full-dimensional potential energy surface for the thermal denitrogenation of 1-pyrazoline using permutationally invariant polynomials neural network (PIP-NN) to accelerate *ab initio* dynamics.
- Rationalized the use of *K*-means clustering for analysis of molecular trajectories and the use of the distance from the cluster centroid as a metric for pruning trajectory data for PES representation.
- Trained and improved the capabilities of an ML classifier that can predict reactivity from the initial conditions of a trajectory simulation.
- Established that multi-reference corrections to single reference energies could be provided by Δ -ML.

C-C Coupling on Graphyne: Computational Approaches to Enhance Reactivity

May 2024 – Oct. 2025

Summer Internship Project - Supervised by Prof. Kaito Takahashi

IAMS, Taiwan

- Investigated differently sized cut-outs of graphyne to pin-point the role of local effects in non-metallic surface-catalyzed CO₂ reduction reactions (CO₂RR).
- Identified the optimal orientation of an external electric field ($\vec{E}$) for lowering the C-C coupling barrier in CO₂RR. Verified the result with the *Reaction-Axis Rule* proposed by Sason Shaik and others. This was done by tracking the flow of electron density.
- Employed a kinetics study using the Transition State Theory for the identification of the optimal condition pair i.e. (dopant, $\vec{E}$ or no $\vec{E}$) for CO₂RR on the graphyne surface. Unified the results with contemporary plane-wave basis calculations.
- Established a linear dependence of the heat of reaction and activation barrier with the external electric field. Provided a dipole moment and charge model to explain the difference in the steepness of curves obtained from doped and pristine graphyne surfaces.

Denoising of Fluorescence Microscopy Images using Machine Learning

Jan. 2024

Term Project for CS460 - Machine Learning (Link to Poster)

NISER, Bhubaneswar

- Restored fluorescence images degraded by Gaussian-Poisson noise using a two-step ML approach: (i) clustering of pixels using pixel intensity values (ii) focused CNN-based denoising within each cluster. (*Dataset*)

- Demonstrated that clustering has the capability to (i) extract noise exclusively from the images (ii) boost the denoising procedure by improving image quality metrics (PSNR and SSIM).

Ab initio DFT/MD Studies of Alkali Metal Cationic Hydration Spheres

May 2023

Summer Internship Project - Supervised by Prof. Sailaja Krishnamurthy

National Chemical Laboratory, Pune

- Benchmarked DFT functional and basis set combinations for water clusters, optimizing the trade-off between accuracy and computational efficiency. Identified the energetically favorable water arrangements around Li^+ and Na^+ ions. The structures were optimized for cluster sizes ranging from $n = 1$ to 7.
- Carried out Born-Oppenheimer Molecular Dynamics (BOMD) on Li^+ and Na^+ hydrated with up to 3 water molecules at 300K. Analyzed mean squared displacement (MSD) of the ions and ion-oxygen separation over course of the simulation to scrutinize aspects of water-shell mobility.
- Employed Hirshfeld analysis and IR frequency calculations to probe the changes in the electronic structure of the coordinated alkali cation.

PUBLICATIONS

Effect of Electric Field on C-C coupling for Doped Graphyne Surface <u>Sahu, P. R.</u> , Dabsamut, K., & Takahashi, K.	(Manuscript under Preparation) 2025
Comparative Studies on Li^+ and Na^+ Cationic Hydration Spheres <u>Sahu, P. R.</u> , Devasia, G., & Krishnamurthy, S.	(Manuscript under Preparation) 2025

SCIENCE COMMUNICATION ACTIVITIES

A Solution to the Anti-Bredt Olefin Synthesis Problem Paper Presentation (For partial fulfillment of the course requirements for C572 - Frontiers in Organic Synthesis)	Nov. 2024 (Link to Slides)
C-C Coupling on Graphyne: A Theoretical Study Student Seminar School of Chemical Sciences, Departmental Seminar	Sept. 2024 (Link to Slides)
Inter-dimensional Portal-Weaving Capabilities of the Funnel-Spider "High-Pot-Thesis!" Science Activities Club (SAC), NISER	Mar. 2023 (Link to Video)
Directed Panspermia Popular Science Article Kshitij 4.0, NISER Astronomy Club (NAC)	Dec. 2022 (Link to Article)

WORKSHOPS AND CONFERENCES ATTENDED

2025 Autumn Workshop on Nanoscale Physics and Chemistry National Center for Theoretical Sciences (NCTS)	October 2025 Taipei, Taiwan
Machine Learning for the Molecular Sciences (ML4MS24) Poster Presentation Digital University of Kerala (DUK), IIT Kharagpur, NISER Bhubaneswar, TIFR Hyderabad	Feb. 2024 Trivandrum, India
Q-Karyashala Workshop on Quantum Information & Technology Indian Institute of Science (IISc.)	Jun. 2023 Bangalore, India

RELEVANT COURSES

Theory: Quantum Chemistry, Chemical Binding, Chemical Rate Processes, Thermodynamics & Electrochemistry, Theory of Nanomaterials, Molecular Spectroscopy & Group Theory, Physical Methods in Chemistry I & II, Advanced Fluorescence Spectroscopy, Theory of Nuclear Magnetic Resonance (NMR), Machine Learning
Laboratory: Physical Chemistry Lab - I & II, Computational Chemistry Lab, Data Structures & Programming I & II

ONLINE COURSE CERTIFICATIONS

Machine Learning Specialization Coursera Stanford University	Sept. 2023
• Supervised Machine Learning: Regression & Classification • Advanced Learning Algorithms: Neural Networks & Decision Trees • Unsupervised Learning, Recommenders, Reinforcement Learning	
Machine Learning with Python Coursera International Business Machines (IBM)	Dec. 2023

SKILLS

Programming Languages: Python, Shell Scripting
Scientific Libraries: PyTorch, TensorFlow, NumPy, Pandas, Matplotlib
Quantum Chemistry Software Packages: Gaussian16, deMon2k, VASP
Molecular Visualization Software: MOLDEN, Chimera
Documentation: L^AT_EX, Google Workspace, Microsoft Office Suite
Interests & Hobbies: Running, Hiking, Painting, Watching movies

POSITIONS OF RESPONSIBILITY HELD

General Secretary <i>The Free Radicals (TFR), Chemistry Club</i>	Mar. 2023 – May 2024 NISER, Bhubaneswar
- Organized <i>A Series of Unfortunate Reagents</i> - a murder mystery game where participants had to piece together chemistry-themed clues to find the murderer. - Coordinated the involvement of the club for exhibitional chemistry during NISER's Open Day and <i>Prerana</i> 2024 (on the occasion of World Children's Day).	
Core Committee Member <i>LitC, Literary Club</i>	Feb. 2022 - Mar. 2023 NISER, Bhubaneswar
- Organized <i>Khat</i> - a prompt-based letter writing competition during the 2023 edition of <i>Twisha</i>, the inter-college event of NISER. - Organized <i>Fabricate</i> - a prompt-based story writing event during the 2022 edition of <i>Umag</i>, the intra-college event of NISER. - Managed the the loan/return record-keeping for LitC's student-run library, ensuring smooth access to books.	

COMMUNITY INVOLVEMENT

Natural Products: Approaches to their Isolation, Characterization and Value-addition <i>Summer Internship Project - Supervised by Prof. Parthasarathi Das</i>	May 2022 IIT (ISM), Dhanbad
- Collected a variety of Jharkhand's local plant produce for extraction of essential oils, oleoresins, and concretes using distillation and solvent-extraction methods. Characterized the compounds using GC-MS and NMR methods. - Used the isolated products for preparation of test batches of value-added soaps, creams, and oils. The products were used for holding skill-development ventures by Atal Community Innovation Centre (ACIC) for tribal marigold farmers. - Interacted with the incubatees at the Atal Community Innovation Centre (ACIC) for discussing the incorporation of naturally extracted oils, oleoresins and concretes in their products. - Regular involvement in exhibitional chemistry demonstrations for school children organized by ACIC.	

EXTRA-CURRICULAR ACHIEVEMENTS

Inter IISER-NISER Sports Meet (IISM) 2024: Table Tennis, Team 4 th position <i>Organized by Indian Institute of Science Education and Research (IISER), Pune</i>	Dec. 2024 Pune, India
Twisha 2023: Pop goes the Culture! (Inter-college Pop. Culture Quiz) <i>Runners-up</i> <i>Organized by Quizone, NISER's Quizzing Club</i>	Mar. 2023 Bhubaneswar, India
Abhiprajna 2022 (UG National Science Quiz) <i>Finalist</i> <i>Organized by Indian Institute of Science Education and Research (IISER)</i>	2022-2023 Tirupati, India
IGNITE 2022: InQuizitive Chemistry (UG National Science Quiz) <i>Winner</i> <i>Organized by Science Activities Club (SAC), NISER</i>	Apr. 2022 Bhubaneswar, India

SPORTS

Long Distance Running <i>Races Ran in Personal Capacity</i>	2024 - present
Khordha Road Half-Marathon 2025 (Chip Time: 2 hours 13 minutes, Position: 30/102) Centroxy Bhubaneswar City Half-Marathon 2025 (Chip Time: 2 hours 10 minutes, Position: 35/156) SBI Green Half-Marathon 2025 (Chip Time: 1 hour 55 minutes, Position: 53/280) TATA Steel Kapilash Half-Marathon 2025 (Chip Time: 2 hours 11 minutes, Position: 83/266)	
Table Tennis <i>NISER Table Tennis Team</i>	Setup: FH - Donic Coppa Gold JO, BH: Donic Sonex JP 2022-2025
- Member of NISER 3-man team, having represented the institute at national and state level inter-college meets. - Notable performance - 4th position at Inter IISER-NISER Sports Meet (IISM) 2024 in the Men's Team event.	